

Xiaochan Xue, Ph.D.

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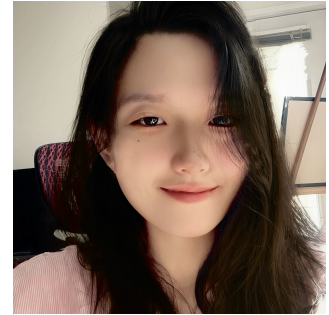
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Summary

I am an **Assistant Professor** in the Department of Electrical and Computer Engineering at the University of Hawai'i at Mānoa. My research lies at the intersection of **wireless communications, cybersecurity, and artificial intelligence**, with a focus on developing **trustworthy AI-native wireless systems**.

My group investigates fundamental and applied challenges spanning physical-layer security, integrated sensing and communications (ISAC), and learning at the radio access network (RAN) edge. Current research interests include privacy-preserving and resilient AI-RAN/O-RAN systems, defenses against adaptive and generative AI threats, intelligent wireless sensing, and Theory-of-Mind (ToM)-inspired network agents for intent-aware, multi-stakeholder decision making.

Recognized for academic and professional achievements, I was awarded the **Excellence Doctoral Fellowship (2023–2024)** and selected as a **2025 NSF CPS Rising Star**. My long-term goal is to advance secure, intelligent, and resilient FutureG wireless systems through the integration of communication, sensing, security, and AI.

Employment History

- 2025 – Present ■ **Assistant Professor.** University of Hawaii at Manoa, College of Engineering, Department of Electrical and Computer Engineering.
- 2024 – 2025 ■ **Lecturer.** Stevens Institute of Technology, Department of Electrical and Computer Engineering.

Education

- 2020 – 2025 ■ **Ph.D., Computer Engineering**, Stevens Institute of Technology
Department of Electrical and Computer Engineering
- 2018 – 2019 ■ **M.S., Electrical Engineering**, Stevens Institute of Technology
Department of Electrical and Computer Engineering
- 2013 – 2017 ■ **B.E., Communication Engineering**, Jilin University
Department of Communication Engineering

Awards and Recognition

Research Awards

- 06/2026 ■ **NVIDIA Academic Grant Award**, NVIDIA Corporation
- 12/2025 ■ **NSF/SaTC Aspiring PI Workshop**, George Washington University (Washington, D.C.)
- 02/2025 ■ **NSF CPS Rising Star**, Vanderbilt University (Nashville, TN)

Awards and Recognition (continued)

Honors and Scholarships

- 2023–2024  **Excellence Doctoral Fellowship**, Stevens Institute of Technology
-  **Graduate Conference Fund**, Stevens Institute of Technology
- 2019–2020  **Research Assistantship**, Stevens Institute of Technology (Master Degree)
- 10/2016  **Third Class Scholarship**, Jilin University
-  **Outstanding Student Leader**, Jilin University
- 10/2015  **Excellent Student Award in College**, Jilin University

Research Publications

Conference Proceedings

- 1 R. Xu, Y. Wan, and X. Xue, “Domusmind: A benchmark for evaluating lifelong smart home agents under drift,” in *ICLR 2026 Workshop on Lifelong Agents: Learning, Aligning, Evolving*, 2026.  URL: <https://openreview.net/forum?id=dCBF23RZYJ>.
- 2 X. Xue, S. Yu, S. Parkar, and J. Li, “Waveform design for ISAC: Trends and future directions,” in *Proceedings of the 2026 International Conference on Computing, Networking and Communications (ICNC)*, to appear, Maui, HI, USA: IEEE, Feb. 2026.  URL: <http://www.conf-icnc.org/2026/>.
- 3 X. Xue, S. Parkar, S. Yu, and Y. Zheng, “AI-assisted Composite ISAC for mmWave Respiration Pattern Recognition,” in *2025 IEEE Annual Congress on Artificial Intelligence of Things (AIoT)*, Accepted, 2025.
- 4 X. Xue, S. Yu, S. Parkar, and Y. Zheng, “ROISD: RIS and O-RAN Assisted Intelligent Sensing for UAV Detection,” in *2025 IEEE Annual Congress on Artificial Intelligence of Things (AIoT)*, Accepted, 2025.
- 5 X. Xue, M. K. Hasan, S. Yu, L. N. Kandel, and M. Song, “Over-the-Air Federated Learning with Enhanced Privacy,” in *ICC 2023 - IEEE International Conference on Communications*, 2023, pp. 4546–4551.  DOI: 10.1109/ICC45041.2023.10278765.
- 6 X. Xue, S. Yu, and M. Song, “Secure Device Trust Bootstrapping Against Collaborative Signal Modification Attacks,” in *IEEE INFOCOM 2023 - IEEE Conference on Computer Communications*, 2023, pp. 1–10.  DOI: 10.1109/INFOCOM53939.2023.10229007.
- 7 X. Xue, S. Yu, M. Song, and C. Xin, “K-Group Random Channel Hopping (K-RCH) Rendezvous for Cognitive Radio Networks,” in *ICC 2021 - IEEE International Conference on Communications*, 2021, pp. 1–6.  DOI: 10.1109/ICC42927.2021.9500643.
- 8 H. Wu, Y. Zou, J. Zhu, X. Xue, and T. Tsiftsis, “Secrecy Performance of Hybrid Satellite-Terrestrial Relay Systems with Hardware Impairments,” in *ICC 2019 - 2019 IEEE International Conference on Communications (ICC)*, 2019, pp. 1–6.  DOI: 10.1109/ICC.2019.8761231.

Journal Articles

- 1 S. Parkar, X. Xue, S. Li, and Y. Zheng, “SPECTRA: Spectrally Disjoint OFDM-FMCW mmWave ISAC with Utility-Driven Mode Selection,” *Manuscript submitted to IEEE Internet of Things Journal (IoTJ)*, under review, 2026, Preprint available on TechRxiv; accessed 2026-02-08.  URL: <https://www.techrxiv.org/users/1024689/articles/1384911-adaptive-mode-selection-in-mmwave-isac-via-spectrally-disjoint-composite-waveforms>.
- 2 M. K. Hasan, X. Xue, S. Yu, and M. Song, “Cooperative NOMA-Based Spectrum Leasing with Multiple Secondary Users,” *IEEE Transactions on Vehicular Technology*, vol. 72, no. 11, pp. 14 543–14 558, 2023.  DOI: 10.1109/TVT.2023.3287230.

Professional Service

- 2026
 - TPC Member, International Performance, Computing and Communications Conference (IPCCC) 2026
 - WCIHS 2026 Workshop Chair, IEEE Healthcom 2026
 - Geo-diversity Co-chair, N²Women Board 2026
 - Demo/Posters Chair, IEEE Healthcom 2026
 - Session Chair, IEEE ICNC 2026
 - TPC Member, IEEE Symposium on Security and Privacy (S&P) 2027
 - TPC Member, IEEE/ACM CHASE 2026
mHealth Security Workshop 2026
 - TPC Member, IEEE Healthcom 2026
- 2025
 - TPC Member, IEEE INFOCOM 2026
ISAC Track
- 2024 – Present
 - **Reviewer**
 - NeurIPS 2026
 - ICML 2026 AIWILD
 - ICLR 2026 AIWILD
 - IEEE INFOCOM 2025–2026
 - IEEE/ACM CHASE 2026
 - IEEE Healthcom 2026
 - IEEE Transactions on Vehicular Technology (TVT)
 - IEEE Internet of Things Journal (IoT-J)
 - AJECE

Institutional Service

- 2026
 - Publicity Committee, Electrical and Computer Engineering, UHM
- 2025
 - Research Committee Member, UHM

Professional Membership

- The Institute of Electrical and Electronics Engineers (IEEE)
- The IEEE Communications Society (IEEE ComSoc)
- Association for Computing Machinery (ACM)

Presentations and Seminars

Presentations

- 06/2021  X. Xue, S. Yu, M. Song and C. Xin, "*K-Group Random Channel Hopping (K-RCH) Rendezvous for Cognitive Radio Networks*", virtual due to COVID. (Invited oral presentation on conference paper)
- 05/2023  X. Xue, M. K. Hasan, S. Yu, L. N. Kandel, and M. Song, "*Over-the-Air Federated Learning with Enhanced Privacy*", Rome, Italy. (Invited oral presentation on conference paper)
-  X. Xue, S. Yu and M. Song, "*Secure Device Trust Bootstrapping Against Collaborative Signal Modification Attacks*", New York area, USA. (Invited oral presentation on conference paper)
- 08/2024  X. Xue, S. Yu and M. Song, "*Secure Device Trust Bootstrapping Against Collaborative Signal Modification Attacks*", 1st Symposium on Emerging Topics in Networks, Systems, and Cybersecurity, Stevens Institute of Technology, NJ, USA. (Poster presentation)
- 02/2025  X. Xue, S. Yu, "*AI-Driven Integrated Sensing and Communication (ISAC) in AI-RAN/O-RAN: Scalability, Privacy, and Security*", NSF CPS Rising Star Workshop, Nashville, TN, USA. (Poster, 17% acceptance rate)
- 04/2025  X. Xue, S. Yu, "*AI-Driven Integrated Sensing and Communication (ISAC) in AI-RAN/O-RAN: Scalability, Privacy, and Security*", Stevens Institute of Technology ECE Department Research Expo, NJ, USA. (Poster presentation)

Seminar

- 02/2025  X. Xue, "*Enhancing Security and Privacy in Distributed Wireless Networks Through Physical Layer Techniques*", Stevens Institute of Technology, NJ, USA. (Invited oral seminar)
- 12/2025  X. Xue, "*Integrated Sensing and Communication in FutureG Systems*", Yeshiva University, NY, USA. (Invited oral seminar)

Teaching Experience

- 2026 Fall  **Instructor.** University of Hawaii at Manoa
Course: ECE 362: Discrete Math for Engineers.
- 2026 Spring  **Instructor.** University of Hawaii at Manoa
Course: ECE 669: Wireless and Mobile Security.
- 2025 Fall  **Instructor.** University of Hawaii at Manoa
Course: ECE 693E: Intelligent and Secure Wireless Networks.
From Wireless Fundamentals to intelligent, secure, and programmable AI-driven Radio Access Networks.

Teaching Experience (continued)

- 2024 – 2025 ■ **Instructor.** Stevens Institute of Technology (During Doctoral Degree)
Course: Information System Security (Graduate Level, On-site & WebCampus).
Designed and delivered course content covering cybersecurity fundamentals, network security, cryptography, and system vulnerabilities.
Developed hands-on assignments and projects to help students understand secure communication protocols, authentication mechanisms, and emerging threats.
- 2019 – 2022 ■ **Teaching Assistant.** Stevens Institute of Technology (During Doctoral Degree)
Course: Applied Machine Learning (Graduate Level).
Supported course instruction by designing assignments, mentoring students during office hours, and guiding them through supervised/unsupervised learning and neural network techniques.
Course: Engineering Design IV (Undergraduate Level).
Instructed this senior design laboratory course, mentoring undergraduates in developing design skills and engineering judgment through design projects in communication, control systems, signal processing, and MATLAB-based simulations.
- 2018 – 2019 ■ **Grader.** Stevens Institute of Technology (During Master's Degree)
Course: Applied Machine Learning (Graduate Level).
Course: Linear System Theory (Graduate Level).

Research Experience

- 01/2022 – 08/2025 ■ **Research Assistant.** Stevens Institute of Technology, AISecLab
Advisor: Prof. Shucheng Yu
- Project: Physical Layer Security in Wireless Networks for Distributed Devices
– Developed a novel trust bootstrapping protocol to mitigate collaborative RF signal modification attacks without additional hardware.
– Designed lightweight, scalable PLS-based confidential computing techniques for federated learning (FL) and multi-device collaboration, including a pairwise cancellable random artificial noise (PCR-AN) scheme for AirComp-based FL.
– Conducted theoretical analysis and experimental validation to ensure security, privacy, and FL convergence.
- Project: mmWave Sensing & Integrated Sensing and Communication (ISAC)
– Built an AI-driven sensing system for contactless breathing pattern detection using mmWave FMCW radar, achieving high real-world detection accuracy.
– Implemented OFDM-based respiration sensing and explored neural network integration for data processing and pattern recognition.
- Project: Wireless Sensing in O-RAN with Security & AI (Ongoing)
– Designing wireless sensing applications for distributed learning under O-RAN.
– Developing Generative AI-integrated ISAC in AI-RAN/O-RAN to enhance cyber-physical systems.
- 05/2019 – 12/2019 ■ **Research Assistant.** Stevens Institute of Technology
Advisor: Prof. Min Song

Research Experience (continued)

- Project: Spectrum (Channel) Utilization Optimization
 - Designed and implemented K-RCH, a group-based random channel hopping protocol for rendezvous in Cognitive Radio Networks.
 - Improved spectral efficiency, reduced time-to-rendezvous (TTR), and enhanced throughput in dynamic spectrum environments.

06/2018 – 08/2018

- **Visiting Researcher.** Nanjing University of Posts and Telecommunications
Advisor: Prof. Yulong Zou
- Project: Secrecy Performance of Hybrid Satellite-Terrestrial Relay Systems
 - Analyzed secrecy outage probability under hardware impairments (HIs) in hybrid satellite-terrestrial relay systems.
 - Derived closed-form results and investigated impacts of relay location, eavesdropper position, and transceiver impairments on secure communication performance.

Student Advising and Mentoring

Ph.D. Advisor. *Department of Electrical and Computer Engineering, UHM*

○ **Saurabh Parkar**

2026–Present

Advisor for research on O-RAN, AI-native wireless systems, integrated sensing and communications (ISAC), and wireless cybersecurity.

Ph.D. Committee. *Department of Electrical and Computer Engineering, UHM*

○ **Ziyi Zhuang**

Fall 2025

Qualifying Examination Committee.

Undergraduate Research Mentor. *Department of Electrical and Computer Engineering, UHM*

○ **Tomohiro Yamada**

Summer 2026

Mentoring on edge AI systems, robotics operation, system logging, and wireless/network performance documentation.

JESSE Summer Intern Mentor. *Department of Electrical and Computer Engineering, UHM*

○ **Kensei Narahara**

2026

Assisted with O-RAN testbed deployment, software installation, experiment documentation, and system validation.

○ **Tim He**

2026

Participated in wireless sensing, AI-enabled networking experiments, and performance evaluation.

○ **Jayden Lau**

2026

Supported data collection, laboratory operation, and dissemination of research outcomes.

Graduate Mentor. *Department of Electrical and Computer Engineering, Stevens Institute of Technology*

Supervised and mentored M.S. students in Professor Shucheng Yu's research group.

○ **Jiapeng Xiao**

Spring 2021

Guided a one-semester master's research scholarship; designed projects and trained him in USRP SDR operation for hardware-software integration.

○ **Zhifan Jiang**

2022–2023

Designed a one-year project on LabVIEW and TMYTEK mmWave antennas; advised research idea development and report writing. Recipient of the *Excellent Research Award*.

◦ **Beula Jose**

2023

Directed a one-semester project on adversary experiments using USRP SDR; enhanced research skills and project performance.

◦ **Ruoxi Li**

2024–2025

Introduced communication systems research using TMYTEK mmWave antennas and OFDM transmission in LabVIEW; advised CSI-based breathing signal collection and data analysis. Subsequently secured a Ph.D. offer.

◦ **Saurabh Parkar**

2024–2025

Supervised projects on O-RAN with POWDER and srsRAN using USRP SDR; trained for presentations and technical reporting. Recipient of the *1st-rank Department Scholarship*. Co-developed projects on mmWave FMCW sensing and 5G NR OFDM respiration monitoring.

Team Mentor. *AFRL Software Defined Radio (SDR) Challenge*

◦ **Team Mentor 1**

2024–2025

Mentored undergraduate and high-school students in developing wireless communication projects using USRP SDR platforms, including MIMO implementation in MATLAB.

◦ **Team Mentor 2**

2025–2026

Led a collaborative team involving Stevens Institute of Technology and Yeshiva University on a RIS-assisted drone detection project using software-defined radio platforms.

Collaborators

Dr. Shucheng Yu, Yeshiva University, USA

Dr. Min Song, Stevens Institute of Technology, USA

Dr. Laxima Niure Kandel, Embry-Riddle Aeronautical University, USA

Dr. Chunsheng Xin, Old Dominion University, USA

Dr. Yao Zheng, University of Hawai'i at Mānoa, USA

Dr. Hanqing Guo, University of Hawai'i at Mānoa, USA

Dr. Shuang Li, University of Victoria, Canada

Dr. Changzhi Li, Texas Tech University, USA

Dr. Yulong Zou, Nanjing University of Posts and Telecommunications, China

Skills

Languages	📖	Strong reading, writing and speaking competencies for English, Mandarin Chinese.
Coding	📖	MATLAB, Python, LabVIEW, GNU Radio, Visual Studio, C/C++, Java, R
Web Dev	📖	HTML, CSS, JavaScript.
Platform	📖	POWDER, COSMOS, open-source srsRAN
Hardware	📖	Software-Defined Radios (SDRs): USRP B/N series, NI USRP 2974 Antennas: mmWave (beamforming), low-frequency (directional, omni-directional) Measurement equipment: Keysight, Rohde & Schwarz
Others	📖	Unity, Final Cut Pro, Studio One, OBS

Relevant Coursework

Wireless & Security	Wireless Communications; Communication Theory; Information Systems Security; Computer Network; Information & Coding Theory.
Machine Learning & Computing	Applied Machine Learning; Algorithms & Data Structures; Computing Principles for Embedded Systems; Computing Principles for Mobile and Embedded Systems; Computer Architecture; Python; C++.
Mathematics & Theory	Advanced Mathematics; Probability Theory & Statistics; Complex Analysis & Integral Transformation; Analytical Methods in Electrical Engineering; Random Signal Analysis.
Circuits & Signal Processing	Digital Signal Processing; Signals & Systems; Data Acquisition; Digital Circuit & Logic Design; Analog Electronic Circuits; High-Speed Circuit Signal Integrity Analysis & Design.
Control	Introduction to Control Theory.